UNSW MARKETING ANALYTICS HACKATHON 2026

# Beyond a Good Story

*When and for whom does persuasive loan language actually speed up funding?*

**Team Cultural Blend**
University of Technology Sydney
Kiva loan data · 1.45 million real loans · 2016–2025

# In 1.45M loans, the story barely registers. The structure carries the signal.

## WHAT PLATFORMS BELIEVE

### “Storytelling is the only lever a platform can coach.”

- Loan amount, sector, and country are fixed at listing.
- Conventional marketing advice suggests adding urgency, mentioning family, or perfecting copy to speed up conversion.
- Naive regressions seem to back this up ($p < 0.001$).

## WHAT 1.45M LOANS REVEAL

### “Structural loan factors beat storytelling by ~10×.”

- Urgency dissolves under country-clustering ($p \approx 0.49$).
- Family framing rests on a 2-country screen illusion ($t(1)=12.7$).
- Loan size, terms, sector, and gender dwarf narrative choices.
- 96.4% of loans pre-disbursed: partner capital waits, not borrower.

## THE STRATEGIC TAKEAWAY FOR KIVA

Do not waste product and marketing resources shipping generic copywriting nudges. Shift platform investment toward structural bottlenecks, partner liquidity cycles, and country-stratified A/B experiments.

# How the 1.45M loan investigation is structured

ACT I · CONTEXT
SLIDES 4–5

## The Operating Reality

*Why narrative advice is pitched, and what really happens on the ground.*

- **The Platform Dilemma:** Text is the only unsealed lever at listing; platforms want to coach it.
- **Market Slowdown:** Same-day funding collapsed from 46% to 30% post-pandemic.
- **Capital Reality:** 96.4% pre-disbursed; partner working capital waits, not borrowers.

ACT II · INFERENCE
SLIDES 6–9

## Empirical Stress-Test

*Testing narrative claims against 1.45M loans with econometric rigor.*

- **Methodological Screen:** 64 of 120 coefficients collapse under country-clustering.
- **Structural Dominance:** Amount, terms, and sector beat storytelling by ~10×.
- **Narrative Scrutiny:** Urgency dissolves ($p \approx 0.49$); family rests on 2-country screen.
- **Topic Modeling:** Topics track loan purpose (water vs farming; 9× swing).

ACT III · STRATEGY
SLIDES 10–12

## The Strategic Playbook

*High-ROI platform decisions, borrower realism, and epistemic bounds.*

- **4-Quadrant Playbook:** Kill generic tips, A/B test family framing, fix structural lag.
- **Borrower Realism:** Set cash-flow expectations; do NOT game gender or story.
- **Scientific Bounds:** Observational associations, completed-loans survival truncation.

APPENDIX · AUDITS
SLIDES 14–17

## Methodological Audits

*Full numerical proofs and statistical architecture for defense.*

- **Two-Country Problem:** The conservative $t(1) = 12.7$ heuristic screen in full.
- **Predictive vs Causal:** Why SHAP importance cannot certify policy rules.
- **The ~10× Comparison:** Decile monotonicities and correlation breakdown.
- **24h Early Warning:** LightGBM classifier architecture (ROC AUC 0.8997).

**PORTFOLIO METHODOLOGY & REPRODUCIBILITY STACK · 1,453,846 REAL LOANS (2016–2025)**

Econometric OLS Duration Modeling · Binomial GLM · Country-Clustered Standard Errors · Conservative Few-Cluster $t(1)=12.7$ Screen · NMF Topic Extraction (8 Components) · LightGBM Predictive Benchmark (Strict Chronological Split, Zero Future Leakage)

# Why platforms coach story: the only lever unsealed at listing

### 1. THE NARRATIVE QUESTION

When a loan's story leans on family, competence, or urgency, does it get funded faster?

### 2. THE MODERATION QUESTION

Does the result shift by WHO the borrower is (gender, geography) and WHERE & WHEN they posted?

### WHY IT MATTERS TO PLATFORM MARKETING

Loan story framing is the one lever a platform can actually coach. Loan size, sector, country, and partner terms are fixed at listing — narrative choices are the only active conversion lever.

# A slower market is the operating reality — and it is partner capital that waits.

## 46.0% → 30.3% → 30.0% funded within 24 hours

- Pre-pandemic, almost half of all loans funded within a day.
- Since 2020: under a third — and four years on, still not recovered (−0.3 points recovery).
- 589,823 loans before vs 565,474 after — not a thin sample, and the slowdown persists to date.

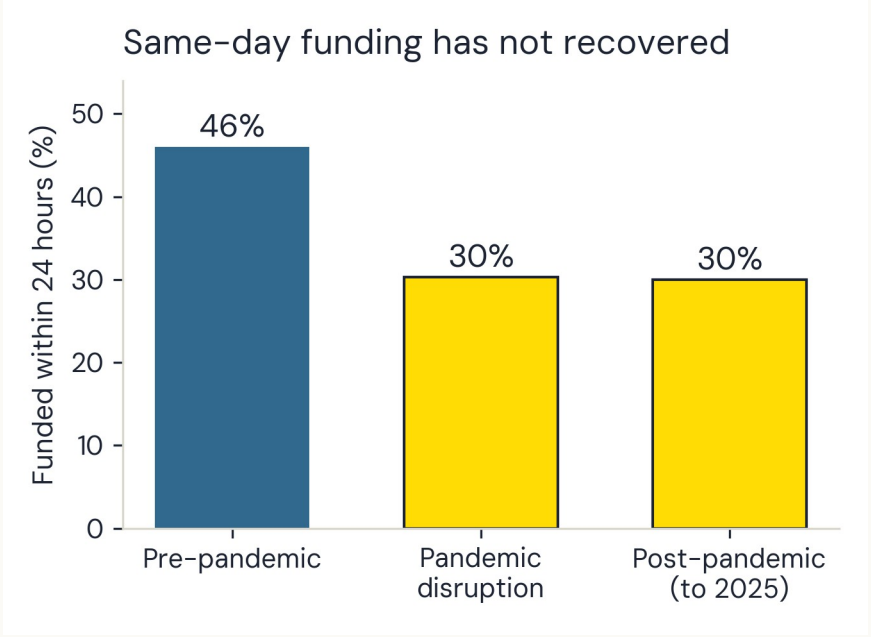



**Figure 2.** Share of loans funded within 24 hours, by analysis period.
Notes: The drop is 16 points; the 'recovery' is −0.3. Exact shares 46.0% → 30.3% → 30.0% (589,823 / 298,549 / 565,474 loans).

**THE OPERATIONAL REALITY · 96.4% PRE-DISBURSED**

**Day 0: Cash Disbursed → Day 24: Kiva Listing → Day 27✦: Reimbursed**

Borrowers do NOT wait for crowdfunding — field partners disburse upfront. A slower market means field partner working capital waits longer to replenish.

# Why big data lies without clustering: how we stress-tested our claims

## 64 of 120

significant coefficients lost significance

- Chronological Holdout: Trained strictly on loans up to 2023; 278,887 loans from 2024–2025 sealed until evaluation. Zero future leakage.
- Country-Clustered Standard Errors: Standard errors clustered across 80+ countries to remove artificial significance inflation from massive sample sizes.
- Stricter Few-Cluster Screen: Where regional claims rested on only 1–2 countries, required the conservative t(1) critical value of 12.7 (not 1.96).
- Role of Machine Learning: SHAP importance used as complementary predictive validation only; cannot certify causal signs or policy rules.

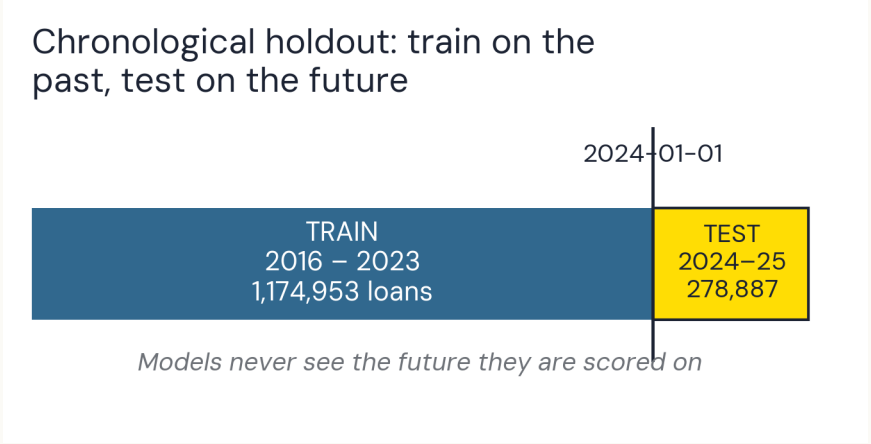



**Figure 1.** Chronological train/test split.
Notes: The 278,887 test loans stayed sealed during training — the models never see the future they are scored on.

# Structural loan factors beat storytelling by an order of magnitude

## 2.3 vs 7.7 days

median speed — female vs male loans

- Loan amount + repayment terms: linked to funding speed ~10× more strongly than any single narrative choice. Bigger ask, longer on the platform.
- Sector alone spans well over an order of magnitude in speed (0.9–12.1 days); country gaps are wider still — 0.2 to 20.5 days.
- Loans posted by women fund in a median 2.3 days; by men, 7.7. None of this is causal — but it dwarfs anything the words do.
- Programme label (whySpecial) spans 0.2 to 30.9 days across 141 well-populated levels — institutional design dominates.

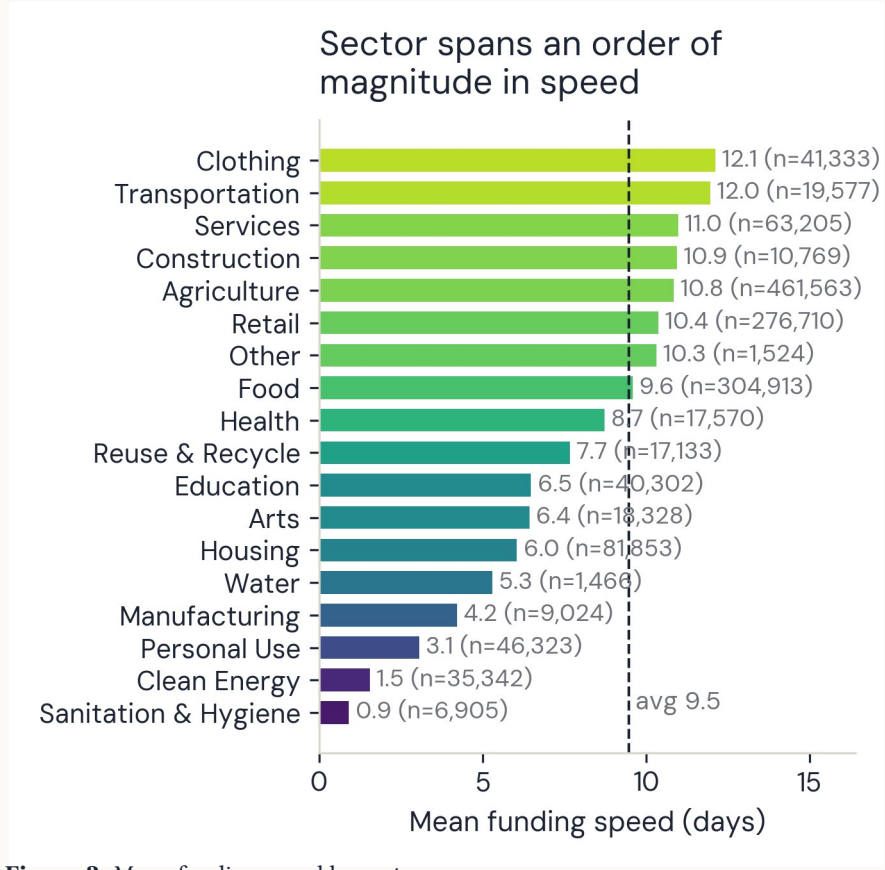



**Figure 3.** Mean funding speed by sector.
Notes: Sanitation loans fund ~13x faster than Clothing (0.9 vs 12.1 days). Counts verified vs executed notebook.

# Storytelling coaching: no robust evidence it moves the needle.

**No narrative result is robust enough to support a writing recommendation**

- Urgency: looked universal (p<0.001) — collapses under country clustering (p≈0.49). Not recommended.
- Family: corrected within-region tests in Middle East (Palestine+Yemen) & Central America (Honduras+Nicaragua) same direction in every fit — but 2 countries each; few-cluster t(1) screen bar is 12.7, not 1.96: p≈0.06–0.12. A hypothesis, not a finding.
- Sentiment: positive tone <-> slower funding, but significance flips between specifications — genuinely open.
- SHAP: no family / agency / urgency feature in the model's top 15 — complementary evidence.

Within-region family-framing slope (duration model, notebook fit): none pass the screen

| region (ctys) | loans | slope | cl. p | few-cl. p | verdict |
|---|---|---|---|---|---|
| Africa (27) | 610,368 | -0.010 | 0.554 | t(26) 0.559 | n.s. |
| Asia (12) | 738,191 | +0.034 | 0.054 | t(11) 0.080 | n.s. |
| C. America (2) | 59,391 | -0.062 | <0.001 | t(1) 0.065 | norm-ref only |
| Middle East (2) | 14,946 | -0.124 | <0.001 | t(1) 0.075 | norm-ref only |
| N. America (1) | 7,559 | +0.011 | 0.062 | not estimable | 1 country |
| Oceania (4) | 23,385 | +0.016 | 0.631 | t(3) 0.663 | n.s. |

**Figure 4.** Average within-region family-framing slope (duration model).
Notes: Against the few-cluster screen (t(1) critical value 12.7), nothing clears it. Authoritative values in analysis_summary.json.

# The themes come down to loan purpose, not persuasion.

## 1.5 → 13.5 days

mean speed across topics — ninefold swing

- Topic modeling on descriptions (TF-IDF + NMF, 8 topics — not keyword counts) surfaces coherent themes: sanitation, clean water, pig raising, family business, smallholder farming.
- The largest single gap anywhere in the analysis: clean water funds in under 2 days; group farming loans take ~2 weeks.
- A theme mostly encodes what the loan is FOR. Lenders prioritize clean water cause, not because the borrower wrote better prose. Structure again, not persuasion.

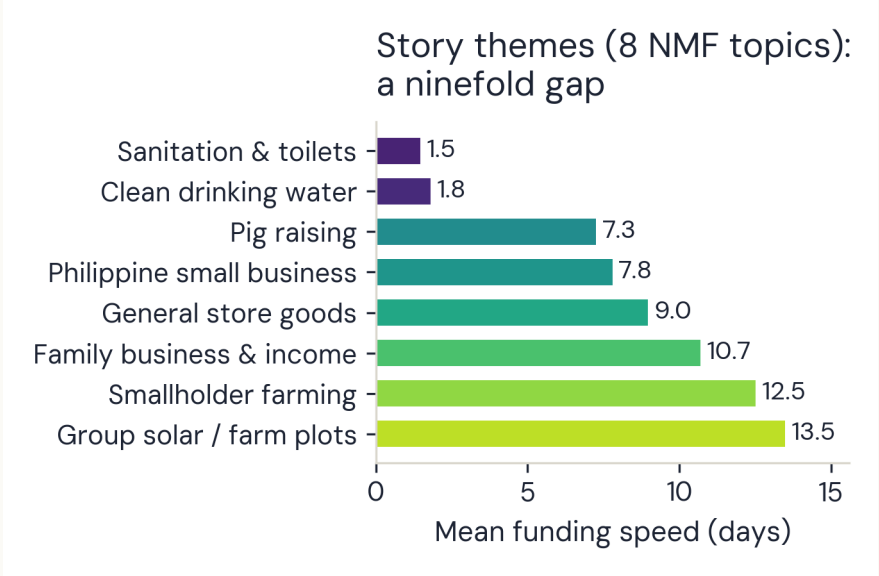



**Figure 5.** Mean funding speed by story theme.
Notes: Water and sanitation stories fund in under 2 days; group farming loans take ~2 weeks — themes track what the loan is FOR.

# What this means in practice for the Kiva team: a 4-action playbook

### DON'T SHIP · Generic Writing Tips

A platform-wide 'add urgency' nudge would be built on unclustered statistical noise ($p \approx 0.49$). Withhold writing coaching to save engineering resources and avoid confusing borrowers.

### A/B TEST · Targeted Family Framing

Run a country-stratified A/B test strictly in the 4 candidate markets: Palestine, Yemen, Honduras, Nicaragua. Randomize prompts at listing; evaluate causal funding speed country-by-country.

### REVIEW STRUCTURE · The ~10× Levers

Review structural friction: Programme label (whySpecial, 0.2–30.9 days), sector, region, and gender. These structural factors drive ten times the variance of any wording effect.

### PROTOTYPE & AUDIT · 24h Classifier

The 24h LightGBM classifier achieves AUC ≈ 0.90 (accuracy 0.840) among funded loans. Before operational deployment, retrain on all listings (including expired and withdrawn).

⚡ **OPERATIONAL REALITY: 96% of loans are disbursed before listing — funding speed is the partner's capital cycle, not borrower waiting.**

# What to expect when listing: setting expectations, not gaming levers

**FOR BORROWERS & FIELD PARTNERS: WHAT TO EXPECT — NOT WHAT TO GAME**

**PROFILE & GENDER** · Loans posted by women fund in a median 2.3 days vs 7.7 male-posted.
**A gap for Kiva to investigate, not a lever for borrowers. DO NOT manipulate who represents the household.**

**LOAN SIZE & TERMS** · The two strongest speed predictors (linked ~10x more strongly than words).
**Amounts are partner-set against real business capital need. Plan cash flow around them; DO NOT shrink the ask.**

**SECTOR & CAUSE** · Speed varies ~13x by sector (0.9 to 12.1 days; water <2d vs farming ~2wks).
**Category is dictated by loan purpose. Set realistic expectations by sector; DO NOT misclassify category to chase speed.**

Note: A bigger ask sits on the platform longer (~2 vs ~19 days). Because 96% of loans are pre-disbursed, this affects partner liquidity cycles, not borrower survival.

# What this can't tell us: three boundaries on the evidence

### 01 · ASSOCIATION, NEVER CAUSATION

**No borrower was randomly assigned a writing style.** All results are observational associations. Narrative features correlate with geographic, economic, and institutional realities that cannot be causally attributed to the borrower's text alone.

### 02 · COMPLETED FUNDRAISES ONLY

**Measures how fast a FUNDED loan funds — not whether it gets funded at all.** The dataset contains completed fundraisers (1,452,203 funded loans). Expired or withdrawn campaigns never enter the platform data, creating survival truncation.

### 03 · TRANSPARENT LEXICAL BOUNDS

**Framing measured with transparent, simple rules — not every nuance of persuasion.** Lexicon word-rate frequencies and sentiment compound scores capture broad stylistic patterns, but cannot capture deep storytelling craft, emotional subtlety, or cultural context.

“In this data, the story barely registers. The structure carries the signal.”

**Thank you — questions.**

Team Cultural Blend · UNSW Marketing Analytics Hackathon 2026

A1

# The two-country problem, in full

- Pooled categories: Middle East = Palestine + Yemen (14,946 loans); Central America = Honduras + Nicaragua (59,391); North America = Haiti alone.
- Few-cluster screen p (duration / 24h / notebook fit): ME 0.12 / 0.21 / 0.08 · CA 0.06 / 0.14 / 0.07 — same direction in every fit, none significant; t(1) critical value 12.7.
- North America: single country — between-country uncertainty not estimable at all.
- The screen is a conservative heuristic: it can downgrade a claim, never certify one. Next step: country-stratified A/B test in those four markets.

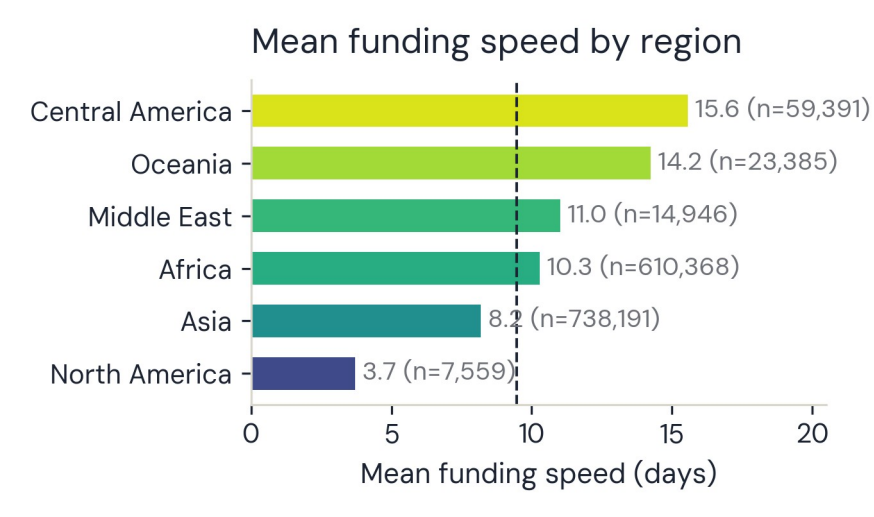



**Figure 6.** Mean funding speed by region.
Notes: North America is Haiti alone (n = 7,559) — fastest region, and exactly why single-cluster inference is off the table.

A2

# Predictive weight ≠ statistical robustness

- SHAP importance from the boosted forecasting model: structure fills the top ranks (amount, term, period, size band, sector, gender).
- No family, agency, or urgency feature reaches the top 15; sentiment is 11th despite its disputed significance.
- Complementary predictive evidence only — a different model, no region interactions; it cannot confirm or refute any coefficient's sign or uncertainty.

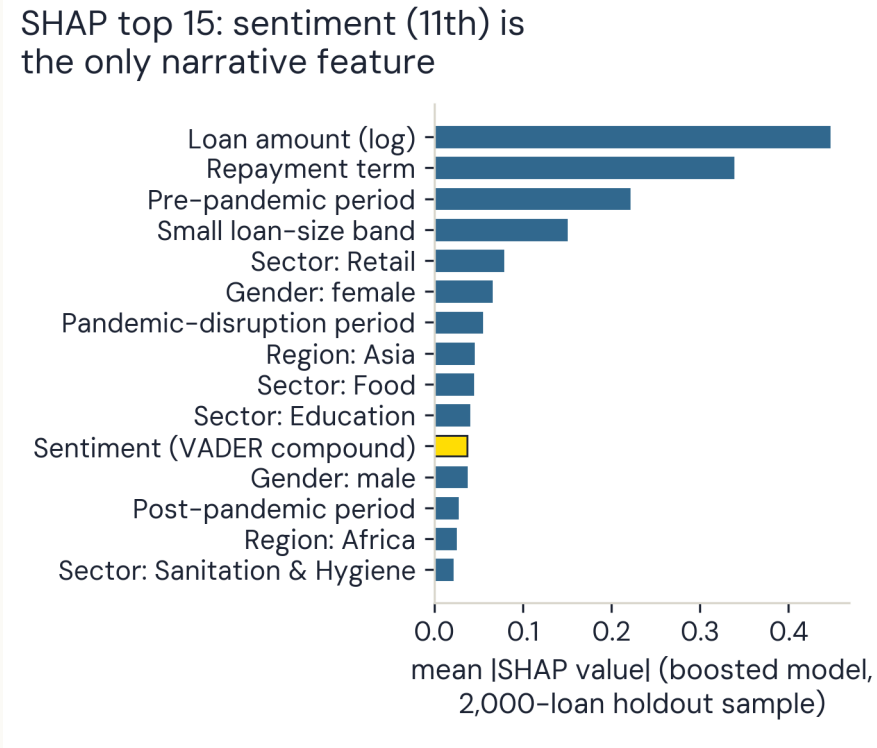



**Figure 7.** Top 15 features by mean |SHAP| value, boosted model.
Notes: Ten structural features outrank the first narrative one. Rebuilt from modeling §8 printed values.

A3

# The '~10x' comparison, exactly

- Correlation with funding speed: loan amount (log) r = 0.43, repayment term r = 0.28 — vs competence 0.058, family 0.019, urgency 0.010.
- Loan-amount deciles also rise almost monotonically (~2 days for the smallest asks, ~18 for the largest) — monotonicity, not the 10x basis itself.
- Gender: female-posted median 2.3 days vs male 7.7 (male coefficient +0.430 survives HC3 AND country clustering, p < 0.0001 — still associational).

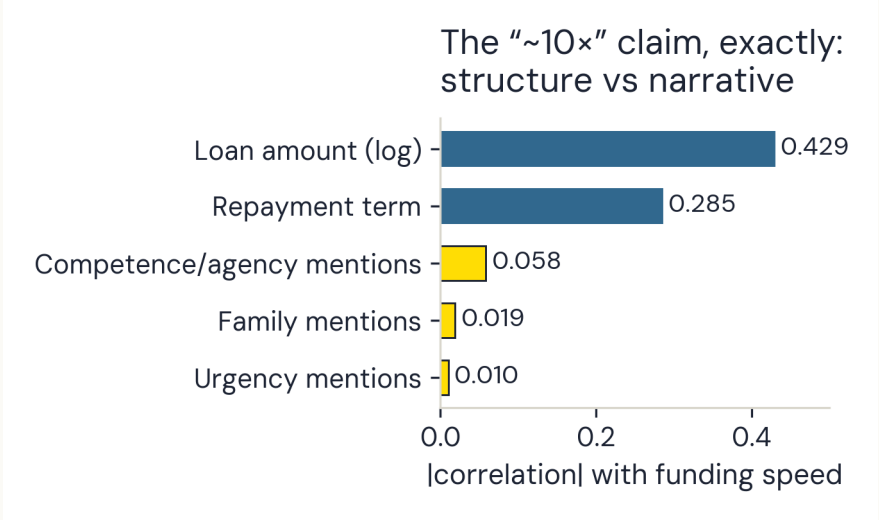



**Figure 8.** Correlation with funding speed: structural vs narrative features.
Notes: Loan amount correlates ~7x stronger with speed than the best narrative signal (0.429 vs 0.058).

A4

# 24h classifier — operating detail

- Holdout (posted 2024-01-01 onward): 278,887 loans; 87,466 funded within 24h vs 191,421 not.
- ROC AUC 0.8997 · average precision 0.8301 · Brier 0.1156 · accuracy 0.840 — structure does the work: no framing feature in the SHAP top 10 (sentiment is 11th).
- Boundary: negative class = 'eventually funded, but not within 24h'. Expired/withdrawn listings never enter the data — a retrospective ranking prototype among eventual funders.
- Path to deployment: all posted listings incl. expired/withdrawn outcomes -> define operational target + censoring window -> retrain, validate -> threshold, calibration, fairness, prospective test.